PG No. 32 $O_{\rm h}$ $m\bar{3}m$ [cubic]

* generator : $2_{001},\ 2_{010},\ 3^{+}_{111},\ 2_{110},\ -1$
* conjugacy class

$[1]:$ 1
$[2_{001}]:$ $2_{001},\ 2_{100},\ 2_{010}$
$[2_{110}]:$ $2_{110},\ 2_{101},\ 2_{011},\ 2_{1-10},\ 2_{-101},\ 2_{01-1}$
$[3^{+}_{111}]:$ $3^{+}_{111},\ 3^{+}_{1-1-1},\ 3^{+}_{-11-1},\ 3^{+}_{-1-11},\ 3^{-}_{111},\ 3^{-}_{1-1-1},\ 3^{-}_{-11-1},\ 3^{-}_{-1-11}$
$[4^{+}_{001}]:$ $4^{+}_{001},\ 4^{+}_{100},\ 4^{+}_{010},\ 4^{-}_{001},\ 4^{-}_{100},\ 4^{-}_{010}$
$[-1]:$ -1
$[\mathrm{m}_{001}]:$ $\mathrm{m}_{001},\ \mathrm{m}_{100},\ \mathrm{m}_{010}$
$[\mathrm{m}_{110}]:$ $\mathrm{m}_{110},\ \mathrm{m}_{101},\ \mathrm{m}_{011},\ \mathrm{m}_{1-10},\ \mathrm{m}_{-101},\ \mathrm{m}_{01-1}$
$[-3^{+}_{111}]:$ $-3^{+}_{111},\ -3^{+}_{1-1-1},\ -3^{+}_{-11-1},\ -3^{+}_{-1-11},\ -3^{-}_{111},\ -3^{-}_{1-1-1},\ -3^{-}_{-11-1},\ -3^{-}_{-1-11}$
$[-4^{+}_{001}]:$ $-4^{+}_{001},\ -4^{+}_{100},\ -4^{+}_{010},\ -4^{-}_{001},\ -4^{-}_{100},\ -4^{-}_{010}$

* symmetry operation

Table 1: Symmetry operations for 3d polar vector.

No.	tag	matrix (polar)	det
1	1	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	1
2	2_{001}	$\begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	1
3	2_{010}	$\begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$	1
4	2_{100}	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$	1
5	3^{+}_{111}	$\begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$	1
6	3^{+}_{-11-1}	$\begin{bmatrix} 0 & 0 & 1 \\ -1 & 0 & 0 \\ 0 & -1 & 0 \end{bmatrix}$	1
7	3^{+}_{1-1-1}	$\begin{bmatrix} 0 & 0 & -1 \\ -1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$	1
8	3^{+}_{-1-11}	$\begin{bmatrix} 0 & 0 & -1 \\ 1 & 0 & 0 \\ 0 & -1 & 0 \end{bmatrix}$	1
9	3^{-}_{111}	$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$	1
10	3^{-}_{1-1-1}	$\begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & 1 \\ -1 & 0 & 0 \end{bmatrix}$	1

continued ...

Table 1

No.	tag	matrix (polar)	det
11	3^{-}_{-1-11}	$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & -1 \\ -1 & 0 & 0 \end{bmatrix}$	1
12	3^{-}_{-11-1}	$\begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & -1 \\ 1 & 0 & 0 \end{bmatrix}$	1
13	2_{110}	$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}$	1
14	2_{1-10}	$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}$	1
15	4^{-}_{001}	$\begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	1
16	4^{+}_{001}	$\begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	1
17	4^{-}_{100}	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix}$	1
18	2_{011}	$\begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$	1
19	2_{01-1}	$\begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & 0 \end{bmatrix}$	1
20	4^{+}_{100}	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}$	1
21	4^{+}_{010}	$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{bmatrix}$	1
22	2_{101}	$\begin{bmatrix} 0 & 0 & 1 \\ 0 & -1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$	1
23	4^{-}_{010}	$\begin{bmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$	1
24	2_{-101}	$\begin{bmatrix} 0 & 0 & -1 \\ 0 & -1 & 0 \\ -1 & 0 & 0 \end{bmatrix}$	1

continued ...

Table 1

No.	tag	matrix (polar)	det
25	-1	$\begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$	-1
26	m_{001}	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$	-1
27	m_{010}	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	-1
28	m_{100}	$\begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	-1
29	-3^{+}_{111}	$\begin{bmatrix} 0 & 0 & -1 \\ -1 & 0 & 0 \\ 0 & -1 & 0 \end{bmatrix}$	-1
30	-3^{+}_{-11-1}	$\begin{bmatrix} 0 & 0 & -1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$	-1
31	-3^{+}_{1-1-1}	$\begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & -1 & 0 \end{bmatrix}$	-1
32	-3^{+}_{-1-11}	$\begin{bmatrix} 0 & 0 & 1 \\ -1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$	-1
33	-3^{-}_{111}	$\begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & -1 \\ -1 & 0 & 0 \end{bmatrix}$	-1
34	-3^{-}_{1-1-1}	$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & -1 \\ 1 & 0 & 0 \end{bmatrix}$	-1
35	-3^{-}_{-1-11}	$\begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$	-1
36	-3^{-}_{-11-1}	$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & 0 & 0 \end{bmatrix}$	-1
37	m_{110}	$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	-1
38	m_{1-10}	$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	-1

continued ...

Table 1

No.	tag	matrix (polar)	det
39	-4^{-}_{001}	$\begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}$	-1
40	-4^{+}_{001}	$\begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}$	-1
41	-4^{-}_{100}	$\begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}$	-1
42	m_{011}	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & 0 \end{bmatrix}$	-1
43	m_{01-1}	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$	-1
44	-4^{+}_{100}	$\begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix}$	-1
45	-4^{+}_{010}	$\begin{bmatrix} 0 & 0 & -1 \\ 0 & -1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$	-1
46	m_{101}	$\begin{bmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{bmatrix}$	-1
47	-4^{-}_{010}	$\begin{bmatrix} 0 & 0 & 1 \\ 0 & -1 & 0 \\ -1 & 0 & 0 \end{bmatrix}$	-1
48	m_{-101}	$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$	-1